Variable Depth $H(x, y)$: Absolute Vorticity $\rightarrow$ Potential Vorticity

Rossby Palooza — Alluvial Rivers & Jet streams

Derivation companion to `numerical/dedalus_meander2/`

The one assumption we drop

The constant-depth model (`dedalus_meander`) assumed a **flat rigid bed**, $H = \mathrm{const.}$ Dropping it (Station 1 assumption (iv)) changes the *conserved quantity*:

$$\text{flat bed:} \quad \frac{D\zeta}{Dt} = \mathcal{C} \qquad \longrightarrow \qquad \text{variable } H: \quad \frac{D}{Dt}\left(\frac{\zeta}{H}\right) = \frac{1}{H}\,\mathrm{curl}\,\mathcal{F}$$

Absolute-vorticity conservation becomes **potential-vorticity (PV) conservation**. The Rossby restoring force gains a **topographic-shear** β. Everything else – jet, erodible banks, low-Froude rigid lid – is kept.

Station 1 revisited: the mass-transport streamfunction

Low-Froude rigid lid, but now $h \approx H(x, y)$ is *not* constant. Continuity

$$\partial_t h + \nabla\cdot(h\mathbf{U}) = 0 \xrightarrow{\mathrm{Fr}^2 \ll 1} \nabla\cdot(H\mathbf{U}) \approx 0$$

So it is the **mass flux** $H\mathbf{U}$ that is non-divergent, not $\mathbf{U}$. Introduce the **mass-transport streamfunction** Ψ':

$$u = -\frac{1}{H}\,\partial_y\Psi', \qquad v = +\frac{1}{H}\,\partial_x\Psi', \qquad \zeta' = \partial_x v - \partial_y u = \nabla\cdot\left(\tfrac{1}{H}\nabla\Psi'\right)$$

The relative vorticity is now a **variable-coefficient elliptic** operator on Ψ' (flat bed: $\zeta' = \nabla^2\Psi'$).

The curl: the PV-tendency equation

Curl of the shallow-water momentum equation (the surface term $-g\nabla\xi$ is a gradient, killed exactly). Using $\nabla\cdot\mathbf{U} = -\frac{1}{H}\mathbf{U}\cdot\nabla H$:

$$\partial_t\zeta' + \mathbf{U}\cdot\nabla\zeta' - \frac{\zeta'}{H}\mathbf{U}\cdot\nabla H = \mathrm{curl}\,\mathcal{F} \quad\Longleftrightarrow\quad \frac{D}{Dt}\Big(\frac{\zeta'}{H}\Big) = \frac{1}{H}\,\mathrm{curl}\,\mathcal{F}$$

The extra **vortex-stretching term** $-\frac{\zeta'}{H}\mathbf{U}\cdot\nabla H$ is what the flat bed did not have. It combines with base advection into a modified β – next slide.

Linearize (base bed $\bar{H}(y)$): the SAME skeleton

Base jet $\bar{u}(y)$ over base depth $\bar{H}(y)$; $\bar{\zeta} = -\bar{u}_y$. The stretching term folds into the advection, and the linear equation is **term-for-term the flat-bed model** with two replacements:

$$\boxed{\partial_t \zeta' + \bar{u}(y)\,\partial_x \zeta' + \beta_{\mathrm{top}}(y)\,\partial_x \Psi' = \mathrm{curl}\,\mathcal{F}', \qquad \zeta' = \nabla\cdot\left(\tfrac{1}{\bar{H}}\nabla\Psi'\right)}$$

$$\beta_{\mathrm{top}}(y) = \partial_y\!\left(\frac{\bar{\zeta}}{\bar{H}}\right) = \frac{2D}{\bar{H}} + \frac{\bar{u}_y\,\bar{H}_y}{\bar{H}^2} \qquad (\text{flat bed } \bar{H} = 1:\ \beta_{\mathrm{top}} \to 2D,\ \zeta' \to \nabla^2\Psi')$$

$\beta \to \beta_{\mathrm{top}}$ (topographic-shear) and $\nabla^2 \to$ the $\bar{H}$-weighted elliptic operator. Flat bed **recovers `dedalus_meander` exactly**.

An honest word on "topographic β"

In a **non-rotating** river ($f = 0$) the base PV is $\bar{\zeta}/\bar{H}$ – relative vorticity only. Hence

$$D = 0 \;\Rightarrow\; \bar{\zeta} = 2Dy = 0 \;\Rightarrow\; \beta_{\text{top}} = 0 \qquad \textbf{(a bed bump alone makes NO } \boldsymbol{\beta}\textbf{)}.$$

The bed *modulates the shear-β*; it does not create a standalone topographic β without planetary vorticity. A formal knob f_0 (default 0) restores true **topographic Rossby waves**:

$$\beta_{\text{top}} = \partial_y\!\left(\frac{\bar{\zeta} + f_0}{\bar{H}}\right) = \underbrace{\partial_y(\bar{\zeta}/\bar{H})}_{\text{shear}} - \underbrace{f_0\, \bar{H}_y/\bar{H}^2}_{\text{topographic}} \qquad (f_0 \neq 0: \text{ waves even at } D = 0).$$

Implementation: well-banded, and why

A survey of the group's Dedalus code found **no variable-coefficient elliptic-LHS precedent**; the one NCC $\times$ unknown on the LHS destabilised the IMEX stepper, and the docs warn that rational $1/H$ NCCs make dense matrices. Safe, precedented route (as in the QG/shallow-water scripts):

- carry ζ' as an **auxiliary field** (first-order reduction);
- multiply the elliptic constraint through by $\bar{H}^2$ to clear the $1/H$:

$$\bar{H}\,\nabla^2\Psi' - \bar{H}_y\,\partial_y\Psi' - \bar{H}^2\zeta' = 0 \qquad (\bar{H} \text{ polynomial} \Rightarrow \text{all-polynomial, well-banded});$$

- multiply the PV tendency through by $\bar{H}^2$ (eigenvalue-preserving row scaling).

The "water outside the channel": which 2nd-order effect?

Drawing the linear field on the *fixed* box $[-1,1]$ while the banks displace by $\eta = \psi_b/U_0$ ($U_0 < 1 \Rightarrow \eta > \psi'$) leaves an O(ampl)-wide strip ("water not filling / water outside"). It is **purely geometric**, *not* a flow. Four second-order things, routinely conflated:

- **Geometric $\delta_0 = ke$** (intrinsic→Cartesian): puts the field IN the meandering channel. **THIS is the gap**; `warp_fill` is its leading order. Parker 1982 keeps $O(\delta_0^3)$.
- **Nonlinear $J(\psi', \zeta') = O(\text{ampl}^2)$**: flow self-advection (fattening/skewing of the bend). Dropped – we are linear.
- **Ikeda $\nu = b/r_0$, kept to $O(\nu^2)$**: the FLOW's response to bend *curvature* (superelevation $\propto F^2$, near-bank velocity), taken at $\delta_0 \to 0$. A *different parameter* – curvature, not amplitude.
- **Secondary flow $A \approx 2.89$**: 3-D helical $\partial u/\partial z$ that scours the outer bank. Depth-averaged away in both models; Ikeda re-imports it as $\eta'/H = -A\mathscr{C}'\tilde{n}$. We omit it (a bed/cross-section effect).

The gap is filled by the coordinate map (warp), not by any flow. Ikeda's $O(\nu^2)$ is curvature-flow at zero amplitude; secondary flow is 3-D bed-scour – both are *different* things, orthogonal to this planform gap. **Both movies** (`multipanel` and `multipanel_eulerian`) therefore render the interior on the warped mesh, so *the two banks are the exact boundary of the flow* – they differ only in scaling: per-frame normalisation (mode shape) vs. one fixed scale (true $e^{\sigma t}$ growth).

Along-channel bed: flow over a wavy bed

A bed varying *along* the channel, $H(x,y) = \bar{H}(y)[1 + a_x\cos(k_b x)]$ (Ikeda's alternate bars), breaks x-homogeneity – Fourier modes couple, so it is **IVP-only** (no normal modes). The base flow is fixed by **discharge conservation** $\partial_x(Hu) = 0$:

$$Hu_0 = \bar{H}(y)\bar{u}(y) \;\Rightarrow\; u_0(x,y) = \frac{\bar{H}(y)\,\bar{u}(y)}{H(x,y)} \qquad (a_x{=}0:\; u_0 = \bar{u},\text{ the } \bar{H}(y) \text{ case}).$$

Flow **speeds over bars, slows over pools**. With prognostic PV $q' = \zeta'/H$ the system stays polynomial-coefficient:

$$H\partial_t q' + \bar{H}\bar{u}\,\partial_x q' - \partial_y\Psi'\,\partial_x\bar{q}_0 + \partial_x\Psi'\,\partial_y\bar{q}_0 + \gamma q' = 0, \quad H\nabla^2\Psi' - \partial_x H\,\partial_x\Psi' - \partial_y H\,\partial_y\Psi' - H^3 q' = 0.$$

Bed↔bank resonance (default $k_b = k$, bar wavelength = meander wavelength = the Ikeda bar↔bend coupling): at $a_x{=}0.25$ the bed shifts σ 0.10→0.11, c −0.26→−0.19 (enhanced growth, slower upstream march).

Verification ladder (all green)

check	what	result
flat-bed reduction	variable-H GEP vs `VL.channel_modes` (H=1)	0.0 **(identical)**
$d3$ EVP vs GEP	flat + bumped bed, both closures	4.4×10^{-9}
topographic Rossby	f_0=0: β_{top}=0; $f_0 \neq 0$: propagating wave	**as predicted**
IVP vs EVP	measured σ, c, flat + bumped bed	0.1%
along-channel bed	`build_ivp_Hxy` at a_x=0 vs $\bar{H}(y)$ EVP	0.1%

Two independent discretisations (Dedalus Chebyshev spectral & an FD+Richardson GEP) agree to $\sim 10^{-9}$ – the $\bar{H}^2$-weighted PV reformulation is correct.

Result & outlook

- A deeper **thalweg** $H(y) = 1 + a_H(1 - y^2)$ shifts the growth band and the **group-velocity flip** k_g (the upstream/downstream momentum boundary) to the right – modestly, because a *symmetric* bump partly cancels in β_{top}.
- **Along-channel bedforms** $H(x, y)$ **done** (discharge-conserving base flow): at bed$\leftrightarrow$bank resonance the bars measurably force the bends.
- **Model is strictly linear**: the $O(\mathrm{ampl}^2)$ nonlinear $J(\psi', \zeta')$ and Ikeda's 3-D secondary-flow scour ($A \approx 2.89$) are *deliberately dropped* (see the four-effects slide) – they are separate physics, orthogonal to the linear channel-Rossby instability studied here.
- Core is one Dedalus driver (`meander_driver.py`); all figures/movies in `postprocessing/`. Flat bed always recovers `dedalus_meander`.